

Pre-feasibility methodology to compare productive uses of energy supplied by stand-alone solar photovoltaic systems: A Tanzanian case study

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ABSTRACT

This paper provides a standardised methodology to assess the suitability of using stand-alone solar photovoltaic (PV) systems for different productive uses of energy (PUE) at a country-wide level. The focus has been on the country of Tanzania, but the methodology is suitable for any location. This analysis has reviewed fourteen different PUE categories, with several sub-categories. The PUE were assessed for a wide range of factors, including cost of the system, potential income from the PUE, and potential market size. This report also highlights the PUE load profiles and energy requirements for comparison.

Assessing a potential productive use is a complex process with multiple stakeholders and decision factors. The analysis here is not meant as a definitive answer, but to highlight the technical characteristics and economics of the potential productive uses for comparison. This methodology is a first pass method to assess the suitability of stand-alone PV systems for PUE applications at a country-wide level. For more detailed analysis of the PUE potential, the enabling environment, socio-cultural context and supporting services must be investigated in much greater detail. The quantitative methodology given here used computer-based simulation tools, including data processing (Excel & Python), geographical information systems (QGIS) and power system simulation (HOMER Pro). Input data from multiple sources, including in-country fieldwork, was used. This analysis has highlighted PUE with smaller energy requirements but low sensitivity to equipment or service cost to be the most suitable for more detailed analysis.

Applying this methodology to Tanzania has shown stand-alone solar PV systems for barbershops, tailors, mobile carpenters, drip-feed irrigation systems and fishing lights as the lowest risk PUE for implementation.

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Introduction

The aim of this paper is to present a multi-parameter first-pass methodology for evaluating and comparing the technical and economic feasibility of PUE supplied by stand-alone solar PV systems.

Access to energy is vital to economic, social, and human development. At present around 770 million people worldwide do not have access to electricity. This is 10 % of the world's population and 550 million of those without access are located in sub-saharan Africa (IEA). Sustainable Development Goal 7 calls for universal access to affordable, reliable, and modern energy services by 2030. Providing a good quality, reliable electricity supply is one method of providing modern energy services which can be used for a wide variety of applications. While grid extension and densification are the least cost options for providing reliable

energy access for around 45 % of the population in Africa gaining access by 2030, mini grids will account for 30 % and stand-alone systems for around 25 % (IEA). In many remote and rural areas, the least cost option to achieve electricity access will be stand-alone solar PV systems due to the declining costs of small-scale solar PV modules (IEA). Access to electricity is not binary, and a multi-tiered approach has been developed by ESMAP which labels six tiers of electricity access from 0 (no access) to 5 (reliable supply) (World Bank Group & ESMAP, 2015). Providing energy supply solutions with suitable financing methods can help people to rise through these electricity access tiers and help break the cycle of poverty.

PUE are “agricultural, commercial and industrial activities depending upon energy services as a direct input to the production of goods or provision of services” (GIZ, 2016). Providing energy access and robust and reliable equipment that mechanises PUE activities – which may otherwise be performed manually – can boost productivity and economic activity, although these effects can take a long time and be costly (Bhattacharyya, 2013). As highlighted in the literature review, there are reports showing the potential benefits of providing different PUE. But

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analysis of the economic viability of PUE systems is typically simplistic without any geographical data or the analysis applies to larger electrical supply systems, such as mini-grids.

The methodology presented here can be used to provide guidance to system integrators and equipment suppliers for preferred viable solutions. To analyse the most straightforward, and hence most economically viable, potential PUE activities, a standard methodology is required. This will help guide equipment and systems suppliers and manufacturers to develop equipment and create supply chains for the most technically and economically viable PUE systems. The more technically and economically challenging PUE systems can then be developed using the knowledge and experience from any initial deployment.

The Republic of Tanzania is used as the example area within this paper, but the methodology is applicable to other locations. Tanzania is an East-African country with a population of 59 million people, of which 37 million people live in rural areas. On average 63 % of the population (37 million) do not have access to a reliable source of electricity (IEA, 2019), which is higher (83 %) in rural areas (Rural Energy Agency, 2017). The Human Development Index for Tanzania, which assesses long-term progress of human development, is 0.538, ranking 154th in the world (United Nations Development Programme). The gross national income per capita is US \$2655 (United Nations Development Programme). The Tanzanian economy is heavily based on agriculture, with 82.6 % of rural Tanzanian households having agriculture or fishery work as their main income generating activity (Rural Energy Agency, 2017). Agriculture accounts for 85 % of Tanzanian exports and 50 % of employment (Water Resources Group, 2013). Over 85 % of the land under cultivation is used by smallholder farmers, typically for subsistence farming (FAO).

This paper will first review work already published, reviewing PUE and investigating the case for access to electricity driving economic development. The methodology is presented, along with the tools required for implementation. Data sources and key model assumptions are given. Results from implementing the model with 14 different PUE scenarios are presented.

Literature review

Providing energy services, such as an electricity supply, to new and more efficient businesses may provide a pathway from electricity access to socio-economic impacts (Peters & Sievert, 2016). Access to electricity alone may not deliver economic and social benefits, however, it can potentially enable firms to better exploit the opportunities presented by changing economic contexts (Fetter & Usmani, 2020). Furthermore, electrification may increase the productivity of activities that do not necessarily require electricity, such as water collection or sewing and it can allow activities that were previously not possible, such as soldering, metalworking or small-scale industrial production which can influence the effective wage for a household (Fetter & Usmani, 2020).

While it is often claimed that there is correlation between electricity access and business or household productivity, and higher income (Futurepump, 2020; Lighting Global, 2019; Morrissey, 2019), there is not always robust evidence for this link (Bhattacharyya, 2013; Grimm et al., 2013; Jeuland et al., 2021). In addition, evidence is scarce for micro and small enterprises operating in the informal sector (Grimm et al., 2013). Looking at grid-based electricity supply, (Neelsen & Peters, 2011) suggests that grid electrification's effects on a business's performance measured by profits are small and, particularly among small manufacturing businesses, the effects of grid electricity on the production process appear to be weak. (Peters et al., 2011) found that, with respect to existing businesses in Benin, there were no clear-cut economic benefits from grid electrification. (Fetter & Usmani, 2020) also found no evidence that grid electrification had any discernible short-term impact on labour-market outcomes in villages in India. Electricity access can also be provided by renewable energy based micro-grids and stand-alone power supplies (Mandelli et al., 2016).

The PRODUSE report (Dombia et al., 2013) found a lack of robust evidence to link electrification and rural micro-enterprise development, although the reduction in cost of solar PV may change those economic assumptions. (Dombia et al., 2013) also highlighted the importance of ensuring robust methodology and reliable evidence.

In some cases, low levels of electricity consumption can be best served by low-cost solar PV stand-alone power supply systems, although there needs to be careful cost-benefit comparisons of on-grid and off-grid solutions (Peters & Sievert, 2016). It remains unclear that off-grid solar can sustainably and affordably deliver the productive use services and increased incomes (Jeuland et al., 2021). Electrification programs that rely on stand-alone technologies must consider the trade-offs between the complexity, the costs, and the benefits of these systems. Larger systems for generating stand-alone solar power may produce broader socioeconomic benefits but generating these benefits may require more expensive interventions (Akin et al., 2017).

As the link between providing modern energy services and economic growth is not well defined there is a need to place efforts into activities that have the best potential to succeed (Jeuland et al., 2021). A rigorous approach to understanding the real-world situation is needed (Fetter & Usmani, 2020). The insights generated can be used to inform spatial targeting of resource-intensive infrastructure by allowing policymakers to better gauge cost-benefit trade-offs and choose appropriate off-grid energy solutions for different contexts (Fetter & Usmani, 2020). Projects should be accompanied by technical and financial assistance to assess productive use potentials (Peters et al., 2011). The methodology presented here is an attempt to provide a first pass assessment of the effort required for implementing various PUE in the specific example of stand-alone solar PV based systems in Tanzania.

The Productive Use Leveraging Solar Energy (PULSE) report (Lighting Global, 2019) reviewed the state of PUE in Africa. Typically, agriculture is the dominant application, with >50 % people having agriculture as their primary household income source. This report reviews the power requirements of a wide range of PUE, but it does not investigate the load profiles nor the energy requirements, which are both crucial to system design.

An analysis of Productive Use of Energy in African Micro-Grids (Booth et al., 2018) focussed on micro-grids in the 5–100 kW range with multiple loads, rather than single-activity PUE systems. Attaching a PUE to a micro-grid system which also supplies domestic loads is sensible from an energy utilisation and economic perspective. Typically, a PV-based micro-grid is under-used during the day and there may be excess power available which could be utilised by a PUE. But micro-grid systems are much more expensive to install and complex to deliver, so the economic situation is very different to supplying single-activity PUE equipment with an integrated power system.

To provide PUE services, there must be access to robust equipment and appliances which are tested to real-world conditions (Kavita Rai et al., 2018). (Peters et al., 2011) found that 68 % of small enterprises in Benin said access to equipment and machines is a major barrier to development, while only 17 % saw access to electricity as a major barrier. Reports on PUE equipment supplied and energy requirements for productive uses with a focus on Tanzania and East-Africa, which inform this analysis, are given in Table 1.

Methodology

This section explains the analysis used to assess the various PUE applications. It highlights the main outputs from the analysis, the processes and calculation tools used and highlights the sources of data. An overview of the PUE applications and the calculation data used is also presented here. Limitations to this analysis are also highlighted.

To generate the methodology presented here, market assessments for stand-alone renewable energy systems for PUE applications have been reviewed, including (Alsop et al., 2017; Eales et al., 2017; Smith

Table 1
Sources of data for PUE equipment.

Title	Ref	Highlights	Data Used
GOGLA: Off-Grid Solar Market Trends Report	(GOGLA, 2019)	Off-grid solar annual market US\$ 1.75 billion	Price ranges of PUE equipment.
Global Leap, GOGLA, World Bank, ESMAP, Coalition Energy for Access: State of the Off-Grid Appliance Market report	(Global Leap, 2020)	Around 3000 solar water pumping systems and 3000 solar refrigeration systems were sold in the six months from January to June 2019. The global off-grid appliance market has the potential to become a US\$ 4.7 billion market by 2020	Penetration of PUE appliances into rural sub-saharan Africa households
Access to Energy Institute: Evaluation of Solar Powered Agricultural Technologies for Productive Use Applications	(Avila, 2019)	Investigated 10 different PUE and supply in detail their assessment of two of these uses: oil extraction and coffee pulping.	Energy cost for diesel-powered off-grid systems.
Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ): Photovoltaics for Productive Use Applications: A Catalogue of DC Appliances	(GIZ, 2016)	Review of the main DC appliances for various PUE applications (water pumping, refrigeration, food processing, other manufacturing).	Basic economic analysis. Indicative prices and power requirements for PUE DC appliances.

et al., 2019; Sumanik-Leary et al., 2016). These assessments typically use strategic indicators such as distance to grid, population density, accessibility, and economic activities, alongside surveys and questionnaires to answer socio-economic questions including current and expected energy use estimation, ability, and willingness to pay, existing and aspirational business activity and other social and demographic indicators. While (Eales et al., 2017; Smith et al., 2019) investigated PUE, but with a focus on Malawi, and (Alsop et al., 2017; Sumanik-Leary et al., 2016) focus on small wind turbines, their methodology was used as the basis of the work presented here. This included using load profiles for the PUE activities and using computational analysis tools, including HOMER for power system design and QGIS for geographical analysis. To improve their methodology, more detailed load profiles, a comparison of energy requirements, and more commercial economic analysis, including company profit margins, is required. An economic sensitivity analysis should also be applied to highlight sensitivity to equipment prices and to the cost of the service. Most market assessment also use in-depth questionnaires and interviews, which produces good data regarding the suitability of PUE activities but are time intensive and expensive. The simplified methodology in this paper aims to help reduce the initial data collection requirements. Similar techno-economic analyses with similar methodology have been performed in Malawi (Sumanik-Leary et al., 2016), Ethiopia (Eales et al., 2016) and Nepal (KAPEG, 2019), although they focus on small wind turbines.

Methodology overview

To assess the technical and economic viability of supplying equipment for different productive uses of energy, the analysis must investigate:

1. The potential income generation and profit from the PUE
This is for the potential entrepreneurs providing the PUE service to local customers
2. The market size for the PUE system
This is for the market actors, supply chain & equipment suppliers
3. The potential risk of investing in the PUE
This is for the potential entrepreneurs & equipment suppliers.

This assessment will require any stand-alone power system to be carefully analysed using typical daily load profiles and any seasonal variations. The economic model for a 'typical' end-user of the PUE must show that additional profits from providing the PUE will cover all the costs of the equipment and power supply system and provide a reasonable profit.

It is assumed that:

- To ensure economic viability, there should be a reasonable demand for the service.
- End use equipment for the PUE should be available, or easy to develop or adapt. For long-term viability, equipment should be robust and high-quality.
- There needs to be local access to any critical business inputs.
- On-going training and business support, local availability of expert advice, low-cost repair and long-term operation and maintenance services will be provided.

It is important to note that this methodology is only one part of any energy delivery model. Practical Action's Participatory Market Systems

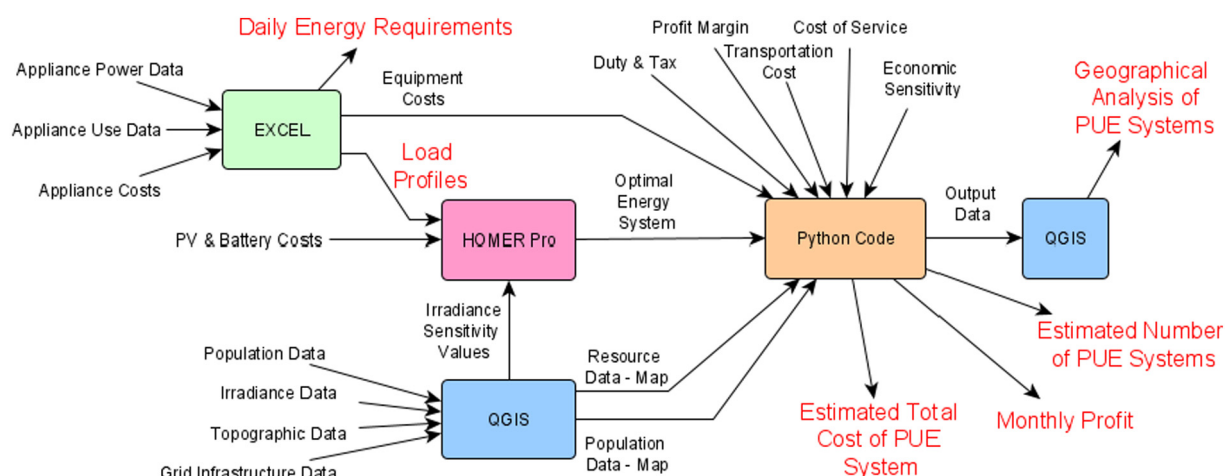


Fig. 1. Overview of analysis process and tools used.

Development framework (PMSD) (Bellanca et al., 2013) and the Energy Delivery Model Toolkit from IIED (Garside & Wykes, 2017) are both multi-disciplinary approaches to energy systems delivery, to which the analysis presented here is just a first step. The ‘enabling environment,’ ‘socio-cultural context’ and ‘supporting services’ should also be mapped out for a full analysis of the market and potential stakeholders.

Analysis process overview

The methodology presented here has several inputs, including geographic, resource, economic and certain socio-cultural factors. It allows the analysis of:

- Power and energy requirements of the PUE activity.
- Technical feasibility and power supply system sizing.
- Geographic feasibility.
- Economic feasibility, including basic analysis of potential income generation and market size.

The standard, quantitative, methodology is shown in Fig. 1, which also highlights the main computing tools used.

Outputs from analysis process

The main outputs from the process are:

Load profiles

Load profiles show how much power is required by the load at a particular time. The load requirements vary on daily, monthly, and seasonal cycles. Load profiles were used to calculate the energy required by the system and hence size the power supply system. Accurate load profiles were required as they could affect either the system reliability, if too small, or the system economics, if too large. From the load profiles, the daily energy requirements for an ‘typical’ user can be calculated (energy is the integral of power over a period of time) and compared. A ‘load factor’ is a percentage of time that the appliance consumes power compared to the length of time the unit is plugged in. For example, a fridge may be plugged in 24 h a day, but the compressor will switch on and off to maintain the temperature, with the load factor of the compressor around 30 %.

Excel has been used to generate the load profile data. Using information from a business survey, along with data from literature sources, the typical appliances used within each PUE were listed along with their average power ratings, load factors and times of operation, typically aligning with the opening times from business survey answers. Using these values, load profiles for each PUE were created.

Daily energy requirements

A Visual Basic for Applications (VBA) macro for Excel was written to produce the year-long hourly load profiles in .csv format. This macro took the hourly power requirements of the PUE created from the PUE appliance lists and load factors. Business survey data also showed which days the PUE operated, and this was used within the macro. The output from the macro was an hour number and an average power requirement for that hour. This can be integrated to find the system energy requirements.

Estimated total cost of PUE system

HOMER Pro (HOMER) is a power supply system optimisation software that takes hourly load profile data (created by the VAB macro), cost parameters (battery costs per amp-hour, solar PV cost per watt-peak and maintenance costs), and solar irradiance data to determine the optimal system design based upon Net Present Cost. HOMER Pro was used for power supply system sizing and economics.

The design criterion used in HOMER Pro was to design a power system to supply the total energy requirement fully throughout the year with solar PV energy. Some of the larger energy users, especially the food cooking vendors, may use fuel stacking, which means other sources of energy are used, such as charcoal for cooking or a diesel generator. This may reduce the power supply system size and hence cost.

The results from HOMER Pro analysis provided the power supply system requirements, including the PV array size and, if required, the battery bank size. The HOMER Pro analysis was run multiple times for multiple different scenarios, by varying the irradiance level values and the cost of the equipment for the PUE. This gave the optimal sized system (including the battery storage size and the PV required) for different irradiance levels, along with the minimum and maximum power supply system costs.

Monthly profit & estimated number of PUE systems

Python 3.8 (Python), a programming language, was used to implement the economic analysis calculations: estimating total system costs, potential market size, and investigate the potential income from each of the PUE activities. Several different ‘typical’ scenarios were created and listed in Table 4. This required several broad-brush assumptions to be made, as the reality of each business and scenario will be much more complex. This analysis is only to rank the various PUE categories for their potential for further, more detailed, analysis. The economic analysis follows the following general approach, but with different parameters used for each PUE scenario.

A ‘pay as you go’ (PAYGo) two-year repayment economic model has been used, with the monthly repayment cost ($Cost_{monthly\ repayment}$) simply calculated from the total system cost divided by 24 months.

The total population using the PUE service is:

$$Population_{using\ PUE} = Population_{without\ grid} \times Percentage_{PUE\ applies} \times Percentage_{using\ PUE}$$

The $Population_{without\ grid}$ comes from geographic population data analysis, knowing the percentages of urban and rural populations without grid connections. The $Percentage_{PUE\ applies}$ is the percentage of people for which the PUE would be suitable. The $Percentage_{using\ PUE}$ is the percentage of the off-grid population who may use the PUE service. The percentage values used for each PUE are given in Table 4.

The number of PUE services per month is:

$$Total\ Services_{month} = Population_{using\ PUE} \times N_{services/month}$$

$N_{services/month}$ is the number of times per month that someone may use the PUE service. For example: for a barber shop, one haircut per month was estimated.

The potential income from the PUE activity can be calculated knowing the cost of the service ($Cost_{service}$), the total number of services provided per month ($Total\ Services_{month}$) and the number of days the business is open ($N_{working\ days}$).

$$Income_{month} = (Total\ Services_{month} \times Cost_{service}) \times N_{working\ days}$$

The potential profit per month is then:

$$Profit_{month} = Income_{month} - Cost_{monthly\ repayment}$$

To assess the maximum potential market size, the number of PUE operations can be calculated, with $Services_{per\ PUE\ per\ day}$ an estimated number of services that can be provided by the individual PUE operation in one working day.

$$Total\ PUE = Total\ Services_{month} / (Services_{per\ PUE\ per\ day} \times N_{working\ days})$$

Using the business survey data, along with reviews of the PUE service and some estimates and assumptions, these values were calculated

Table 2
Overview of the PUE survey questions.

Location of business
Electricity access
Opening hours
Typical equipment used, including power ratings
Approximate number of customers per month
Approximate cost of service
More specific questions asked for different PUE – for example: crops grown by farmers or transportation time for refrigeration of fish

for each of the PUE scenarios. The sensitivity of these economic calculations was assessed by varying the cost of the PUE equipment by 50 % (which would affect the monthly repayments) and the cost of the service by 50 % (which would affect the monthly income). These sensitivities were used to investigate the variation in profit and hence the 'risk' of the PUE. When varying the costs by these sensitivities, if the variation in profit is large, then the PUE service has higher 'risk.'

Geographic analysis of PUE systems

QGIS (QGIS) is a graphical information system (GIS) program for analysing the geographic data. This was used to analyse population data, the irradiance data and, if required, the water table depth data. Data from the HOMER Pro and Python analysis was fed back into QGIS to create maps of potential market size for the PUE.

Business survey

Basic information regarding PUE appliance usage and cost of services was obtained through an informal field-based survey completed in the Mwanza region of Tanzania in February 2020. This survey attempted to interview at least ten businesses for each of the PUE services listed in Table 4, with data from over 130 businesses in total. Businesses were chosen at random by paid local market researchers and conducted in local languages. The main questions asked within the survey are

given in Table 2. The full survey questions and results are available within Appendix 1: Business Survey Data. The results from these surveys were translated and used to create 'typical' user profiles and inform this analysis.

The results from some of these questions were ambiguous as more context or technical knowledge from the surveyor was sometimes required and the translation services were not perfect. But they provided initial values for this analysis, along with an estimate of the variation in the values between the businesses surveyed. These surveys all took place in the northern region of Tanzania, so applying this data to other areas may not be suitable.

Key methodology assumptions

Table 3 gives the other data sources used in this analysis, required for the methodology given in Fig. 1. Estimates of the PUE costs have been made through literature review, discussion with equipment manufacturers and analysis of equipment already on the market. These are given within Appendix 2: Cost of PUE Equipment. The total cost of the full system includes the equipment required for the activity and the solar PV power supply (including the battery bank, if required) to provide the energy. Import duty, company taxation, transportation costs and company profit were then used to calculate the estimated total system cost for the PUE.

PUE applications analysed

A wide variety of PUE applications have been highlighted in (Lighting Global, 2019) and through discussion with a partner organisation, Africa Power Ltd., who supply solar power systems within Tanzania and have knowledge about the local market and their customer's needs. Table 4 shows the different PUE types chosen for analysis and highlights the main assumptions made for each PUE type. This list covers activities already happening within Tanzania, as these have better potential uptake and have local knowledge

Table 3
Data sources for the analysis.

Data item	Data source	Value
Appliance power & usage data	Business Survey GIZ "Catalogue of DC Appliances" (GIZ, 2016) Verasol website (VeraSol)	See Appendix.
Appliance costs	Internet research	See dataset
PV cost	From Africa Power Ltd., the Partner Organisation (PO)	\$0.80/W _p \$2.00/year O&M
Battery cost	From Africa Power Ltd., the PO	\$2.00/Ah 50 % max. depth of discharge \$0.30/Ah/year O&M
Population data	World Pop https://www.worldpop.org/	Map of values (See Fig. 8)
Electricity access data	Energy Access Report – Tanzania https://www.nbs.gov.tz/nbs/takwimu/rea/Energy_Access_Situation_Report_2016.pdf	Map of values (See Fig. 8)
Irradiance data	PV GIS http://re.jrc.ec.europa.eu/pvg_tools	Map of values
Topographical data	Open Street Map www.openstreetmap.org	Map of values
Grid infrastructure data	World Bank Energy Data https://energydata.info/dataset/tanzania-tanESCO-power-systems-2015	10 km buffer distance from power lines to define urban electrified areas.
Import duty – finished consumer goods	Tanzanian Tax Summary http://taxsummaries.pwc.com/ID/Tanzania-Corporate-Other-taxes	25 %
Import duty – solar PV & accessories		0 % (Exempt)
VAT – normal		18 %
VAT – solar PV & accessories:		0 % (Exempt)
Corporation tax		30 %
Profit margin	From Africa Power Ltd., the PO	40 %
Transportation cost	From Africa Power Ltd., the PO	30 %
Cost of service	Business Survey	Various. (See Table 4)
Economic sensitivity	For both the equipment cost and the service cost.	±50 %

Table 4
List of productive uses of energy investigated.

Ref.	PUE name & overview	Employees	Average cost of service	% PUE applies	% Using PUE	Services per person per month	Working days per Month	Services per PUE per day
1	Barber shop Haircutting services for male population.	Small: 1 Large: 3	1800 TZS (\$0.78) per haircut [survey, 2020].	49.9	70	2	26	5
2	Beauty salon Haircutting and beauty process services for female population. Wide variation of service prices from 3000 TZS to 30,000 TZS, typically negotiated with each customer.	Small: 1 Medium: 2 Large: 4	5000 TZS (\$2.16) per treatment [survey, 2020].	50.1	85	2	22	5
				50.1	85	2	22	10
							22	20
3	Tailors Sewing machines used for clothing manufacture, repairs, and alterations. Wide variation of service costs from 1000 TZS to 20,000 TZS.	1	2000 TZS (\$0.86) per job [survey, 2020].	100	80	1 per year	26	5
4	Bar/café/restaurant Equipment required for bars and cafes and entertainment businesses. Three scenarios investigated: 1. Adding a large TV for cinema or sports viewing. 2. Adding a karaoke machine for entertainment. 3. Adding a refrigerator for drinks.	3	These services do not have a direct cost, they are additions which can help to increase income at the venue.	100	50	4	26	43
5	Office services Access to office equipment, such as internet café, printing, and faxing services.	1	1160 TZS (\$0.50) [survey, 2020].	100	70	1	26	10
6	Small manufacturing Local small-scale manufacture of products.	Carpentry Workshop: 3 Mobile 1 Carpenter: 3 Metal Workshop: 3	70,000 TZS (\$30.00) per job [survey, 2020].	100	85	1 per year	26	1
				100	85	1 per year	26	1
				100	85	1 per year	26	1
7	Egg production Increased egg production through additional lighting. Increase in egg yield: 25 % (Jácome et al., 2014; Jacob et al., 2014) Egg incubation (FAO, 2003): 100 egg unit. Egg incubation (FAO, 2003): 500 egg unit. Egg incubation (FAO, 2003): 1000 egg unit.	1 1 1 1	133–300 TZS (\$0.06–0.18) per egg at market (Vera et al., n.d.) \$1.51 per chick [survey, 2020]	100	85	N/A ^a	30	10 eggs
				100	85	N/A ^a	30	5 chicks
				100	85	N/A ^a	30	25 chicks
				100	85	N/A ^a	30	50 chicks
8	Fishing Lights Waterproof lights used to lure fish into nets, for night-time use. This PUE would replace the cost of kerosene for lights typically used. 4–10 lights used per boat (Gengnagel et al., 2013; Ministry of Agriculture Livestock and Fisheries, 2016). 4–8 crew per boat (Gengnagel et al., 2013). Kerosene use 1.25 l per lamp per night (Gengnagel et al., 2013)	Not required.	2120–3200 TZS (\$0.95 – \$1.40) per litre for kerosene [survey, 2020].	100	1.5	N/A ^a	18	N/A ^a
9	Mobile Phone Charging Cell-phone charging kiosks.	1	350 TZS (\$0.15) per phone charge [survey, 2020]	100	50	1 per week	26	11
10	Refrigeration Refrigeration for dairy to reduce the loss due to spoilage. Spoilage rate at present: 50 % [estimate]. Spoilage rate with chilling: 10 % [estimate]. Ice-making for chill fish for preservation during transportation. Diesel icemakers used at present. Refrigerator for a small shop. This would be a benefit for the customer, rather than direct income.	1 3 1	1000 TZS (\$0.43) per litre milk [survey, 2020]. 3000 TZS (\$1.35) per 5 kg ice block [survey, 2020]. 200 TZS (\$0.086) additional charge per drink [survey, 2020].	100	2	N/A ^a	30	40 l daily
				100	1.5	N/A ^a	17	50 × 5 kg blocks
				100	50	8	30	20
11	Food Drying For drying higher-value fruit/vegetable crops. The economic value of this activity is hard to assess. Drying crops usually performed in the open air. Using specialist equipment for controlled drying should reduce the spoilage rate.	Community-Owned. 40 Households benefitting.	100–300 TZS (\$0.12 – \$0.37) per kg (high season crop price) (Match Maker Associates, 2011). 300–500 TZS (\$0.37 – \$0.62) per kg (low season crop price) (Match Maker Associates, 2011).	100	85	N/A ^a	30	1000 kg food per year

Table 4 (continued)

Ref.	PUE name & overview	Employees	Average cost of service	% PUE applies	% Using PUE	Services per person per month	Working days per Month	Services per PUE per day
	Spoilage rate in open air: 34 % (Nonclercq et al., 2009). Spoilage rate with controlled drying: 10 % [estimate]. Mangoes used as example crop for this assessment.							
12	Agri-processing A wide variety of equipment and processes are required. In this analysis maize milling is investigated. Typically, maize is milled with a diesel-powered mill for households when required. 8 kg of maize milled per person per month with 4.8 people per household. 50 households serviced by one mill [estimate].	1	200 TZS (\$0.24) per kg maize [survey, 2020].	100	100	8	26	100 kg maize milled per day.
13	Water Pumping Systems: Two systems investigated. A map of distance to groundwater was used for geographical analysis (Bonsor & Macdonald, 2011). Irrigation for 1 ha of cereal crop production. Maize crop used in this analysis. Flood irrigation is assumed with an efficiency of 55 % (Water Resources Group, 2013). 160 m ³ water required per day (FAO, 2020b). Maize yield at present: 1.4–3 t per hectare (FAO, 2020b). Assumed one additional crop yield produced per year with irrigation.	1	470 TZS (\$0.20) per kg maize (FAO, 2020b).	100	82.6	N/A ^a	30	2000 kg per year sold
	Drip-fed Irrigation of 0.25 ha of market garden vegetable production. Tomato production used in this analysis (Water Resources Group, 2013). Drip feed irrigation is around 95 % efficient (Everaarts et al., 2011; Water Resources Group, 2013). 14 m ³ water required per day (Yara Tanzania, 2019). Tomato yield at present: 5.85–34 t per hectare (Everaarts et al., 2011). Assumed one additional crop yield produced per year with irrigation.	1	670 TZS (\$0.29) per kg tomato (FAO, 2020a).	100	82.6	N/A ^a	30	4000 kg per year sold
14	Cooking – Street vendor: From additional field work (Perrett, 2020), four distinct types of food vendor were investigated: Chips Mayai: This is a Tanzanian dish of fried potatoes with fried egg. Cooked on demand. Rice & Beans: These vendors sell a wide range of foods including rice, beans, ugali, chapatis and meats. Chicken: 4–5 chickens cooked at a time, cooked by boiling in stock. Dagaa Fryer: This is a specialist activity performed on the locally caught small fresh-water fish, Dagaa. These are fried on the beach in large quantities for preservation.	3 3 3 5	1500 TZS (\$0.65) per serving (Perrett, 2020). 1500 TZS (\$0.65) per serving (Perrett, 2020) 3000 TZS (\$1.30) per serving (Perrett, 2020) 3000 TZS (\$1.30) per kg (Perrett, 2020)	100 100 100 100	50 50 50 1.5	13 13 13 N/A ^a	26 26 26 17	60 47 50 150 kg fish

^a See Appendix 3: Services Per Person Per Month Calculation for more detail.

regarding the service. The PUE applications cover agriculture, industrial and commercial markets. Social activities, such as health care, education and water provision were not included, as the economic assessment for social activities will be different.

Limitations to methodology

The methodology presented here is a first pass, quantitative, analysis. Other factors may affect this, such as the research & development required for the PUE equipment and the readiness of potential businesses within the area, along with gender mainstreaming and social inclusion factors.

Other limitations to this methodology include:

- Each PUE system is an average representation – the reality will be far more variable.
- Economic calculations are simplistic.

- Seasonal variation in customer demand is not included.
- The accuracy of the results depends on the ‘in the field’ questionnaire results.
- Government applied duty and transportation costs are single value and will be more variable for each individual PUE example.

Results & recommendations

Load profiles

The PUE overviews, given in Table 4, along with equipment datasheets, were used to create load profiles for ‘typical’ systems, as shown in Fig. 2. Here we define a ‘typical’ system as providing the service provision given in Table 4 with realistic usage assumptions. For example, a ‘typical’ small barber would be one employee

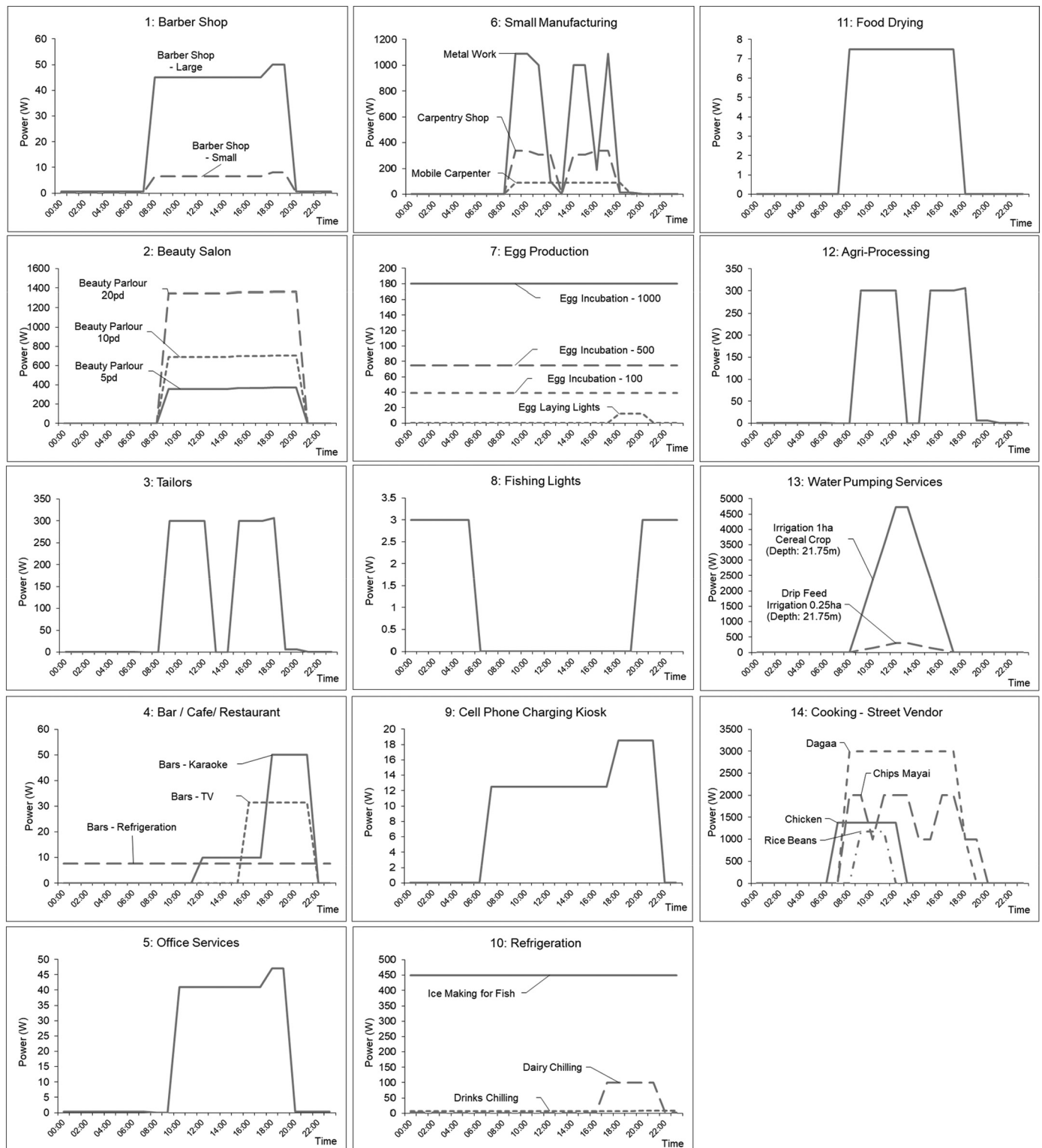


Fig. 2. The 'typical' load profiles calculated & used in this analysis.

providing five haircuts during a working day, using a single set of hair clippers with lighting during the evening. Full details of the values used to create the load profiles are given in the Appendix 4: Load Profile Data.

Note in Fig. 2 the large variation in power requirements between PUE, from <10 W, for fishing lights (Fig. 2, graph 8) and food drying (Fig. 2, graph 11), to over 1500 W, for food cooking (Fig. 2, graph 14)

and water pumping (Fig. 2, graph 13). This large variation in power requirements will affect the power supply system design – it shows there is not a 'one system fits all' solution.

The basic shapes of these load profiles were similar to other analysis performed in (Smith et al., 2019; Williams et al., 2018), although measured data shows higher peaks in the evenings for some of these activities, such as beauty salons (Fig. 2 graph 2). Certain of these PUE load

Table 5
Categories of power supply system.

Category of system	Explanation
Medium power application	These use a battery-based stand-alone PV system. They can provide DC or AC power via an inverter. Typically, between 10 and 500 W provided.
Low power application	These items are very low power (< 10 W) and may have a small integral battery bank, rather than a separate power system.
Refrigeration	Using a form of cooling system. These may or may not have batteries within the power system, as the energy could directly power the cooling system.
Water pumping	These are direct solar PV to water pumping systems with no battery.
Cooking	Electric cooking systems may or may not use a battery bank. These systems would usually be used with fuel stacking and used to offset some of the cooking energy requirements.

profiles were a good match for the solar resource, as the PUE activity takes place during the daytime. This helps reduce the size of any energy storage system and hence reduce system cost.

The PUE activities given in Table 4 were given 'categories' of power systems required, as shown in Table 5. This was only used to help highlight similar PUE activities within this results section, for example: to highlight all the systems providing refrigeration services.

Daily energy requirements

From the load profiles in Fig. 2, the estimated daily energy requirements for each of the PUE activities given in Table 4 were calculated and shown in Fig. 3. A logarithmic scale makes it easier to compare the wide range of daily energy requirements. This graph also overlays

the average energy provided by different sized solar PV modules for average irradiance levels in Tanzania, to help provide context for the data.

Some of these daily energy requirements can be compared to other analysis, shown in Table 6, although comparison data is scarce.

Fig. 3, the estimated daily energy requirements, shows which systems are feasible with smaller PV systems. Typical PV systems provided by the partner organisation are in the range of 20 to 320 W_p, so this graph instantly highlights which PUE may be feasible and which would need a larger, more expensive power system. The smaller systems are typically small LED light units, just requiring 20–100 Wh per day. The larger systems are food cooking system or larger water pumping systems, requiring over 10 kWh per day. It is interesting to note the significant difference between the drip-fed irrigation system, which may be practical and is in the middle range, and the 'flood' irrigation system for cereal which is the largest daily load and potentially expensive to implement. High efficiency refrigeration is a small load, although there may be wide variation from the 'typical' scenario used here.

PUE system cost

The range of estimated cost for the various PUE systems, including the power supply system and the equipment required is given in Fig. 4, again given on a logarithmic scale, to make the data easier to compare.

As would be expected, with lower energy requirements a smaller and hence lower cost system can usually be used. There is variation with cooking systems having a higher cost than the larger energy demand irrigation system. This is due to the additional equipment required for cooking systems, either with an inverter for AC systems or with higher priced DC cooking systems. Irrigation systems also do not need batteries, so their estimated total cost, while still high, is lower than the street vendor cooking systems. Refrigeration systems have a

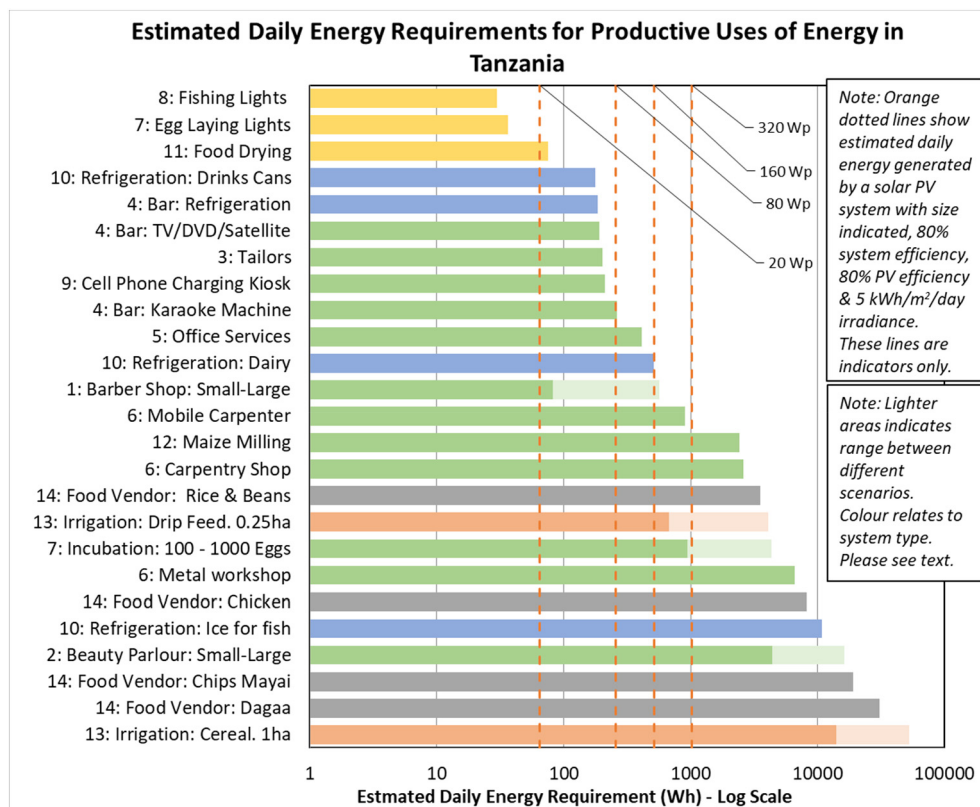


Fig. 3. Estimated daily energy requirements for various PUE.

Table 6
Comparison of daily energy requirements.

PUE	Estimated Daily energy (Wh)	Literature Daily energy (Wh)	Notes
Egg incubation	936–2628	2400 (Booth et al., 2018)	Similar.
Maize milling	2415	20,000 (Booth et al., 2018)	Different. This is due to comparing a small high-efficiency maize mill with a large diesel-powered maize mill. Maize mills come in a wide variety of power ratings & it is difficult to find comparable data.
Beauty parlour	4384–10,324	2600–3100 (Hamidu & Sarbah, 2016)	Similar. Literature data is for Ghana and the number of customers is unknown.

slightly higher cost than their ranking in the daily energy requirements, due to the cost of high-efficiency DC refrigerators. There will be a trade-off between higher cost-efficient products and using lower-efficiency systems with more PV panels, which would be interesting to investigate further.

Fig. 4 is particularly useful for organisations supplying equipment for PUE activities – are these costs practical for the potential market? If not, then implementing these systems will require different funding mechanisms.

Market size, profit and risk

The three main objectives were to investigate the PUE market size, average monthly profit from the PUE activity, and the potential risk of investing into developing this PUE. Using the methodology presented

here, each of the PUE systems were analysed for comparison and results are given in Figs. 5–7.

Fig. 5 shows that systems for the rural farming communities, such as egg laying lights and irrigation systems, have a large market size. This was due to the large farming community, with 82.6 % of Tanzanians giving agriculture or fishing as their main source of income (Rural Energy Agency, 2017).

Fig. 6 gives the range of estimated potential monthly profit from the PUE. The range comes from the sensitivity in the PUE appliance cost (with a 50 % variation in the cost of the PUE equipment) and the service cost (with a 50 % variation in the cost of the service). Any value below zero will mean that there is the potential for that PUE to make a loss in the worst-case scenario and these PUE applications should be considered high risk for any entrepreneur providing these services. The average profit per employee from the PUE activities varies from \$10 to \$1500 per month with a wide range of activities, some of which, such

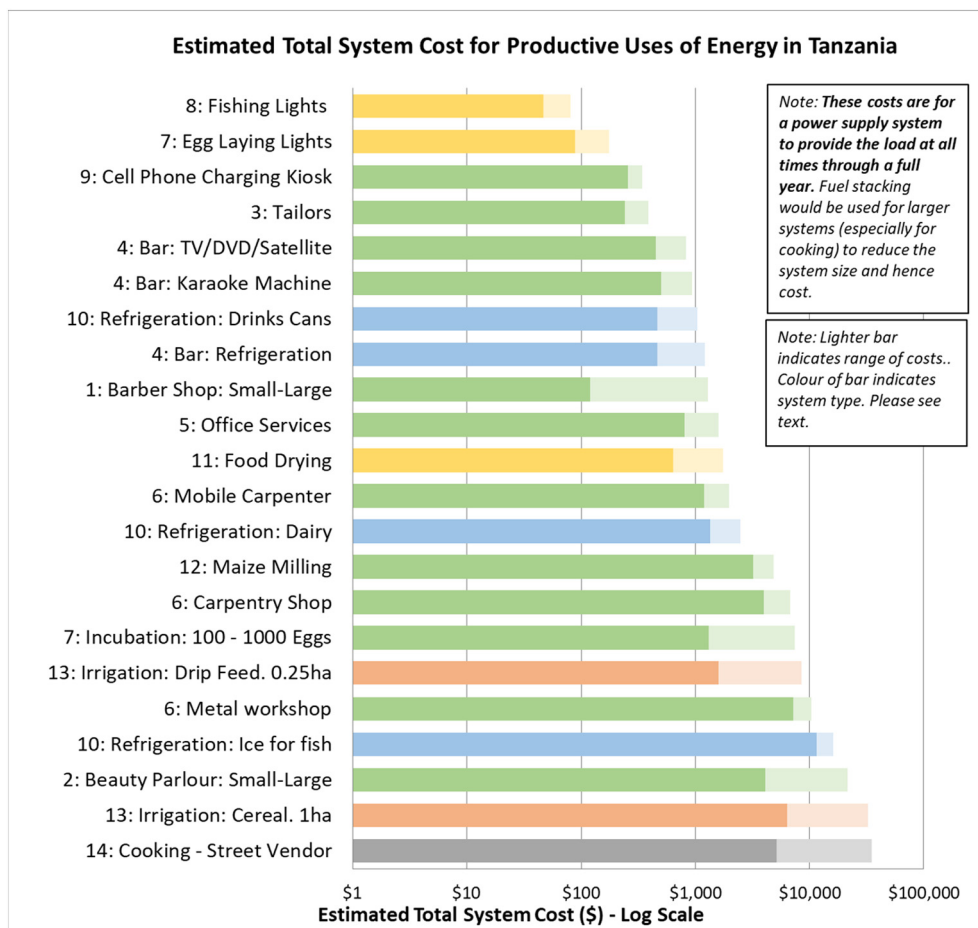


Fig. 4. Estimated total system cost for various PUE.

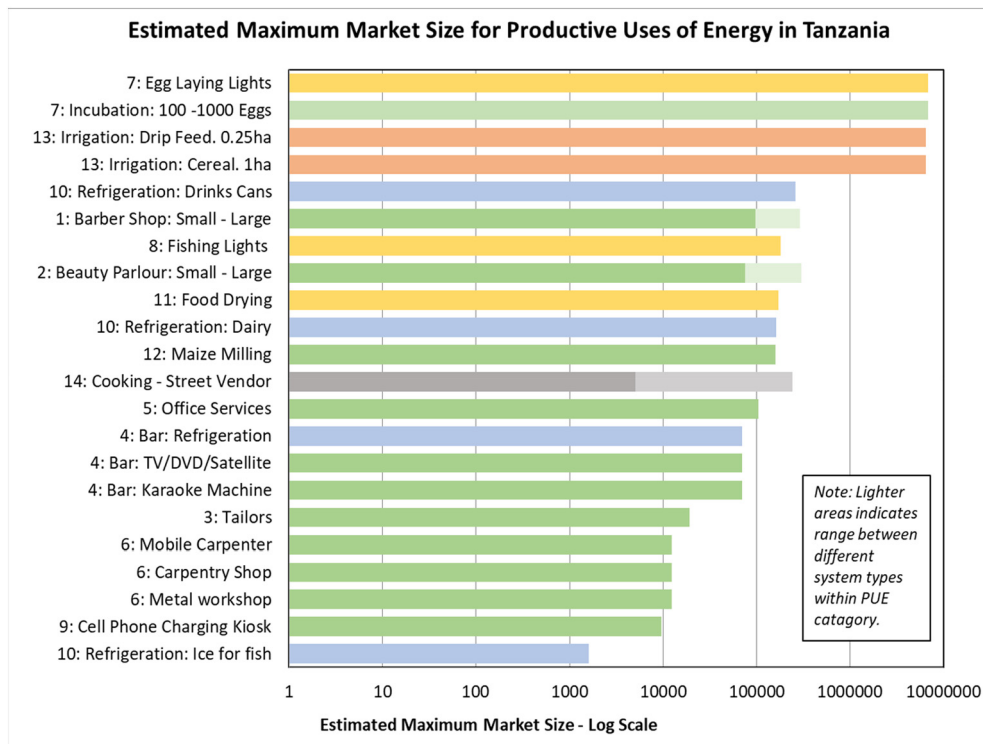


Fig. 5. Estimated market size for the PUE applications in Tanzania.

as egg laying lights, run alongside other economic activity. The average monthly income in Tanzania was \$220 (United Nations Development Programme). Ideally the PUE should provide a reasonable monthly profit, but the range should not go below zero. So drip-fed irrigation, workshop facilities and maize milling provide the best monthly profit.

A 'risk' factor has been determined, which is the minimum monthly potential profit divided by the cost of the system and shown in Fig. 7. This gives a 'risk' indicator per dollar invested, so we can compare different priced systems equally. The lower this value, the greater the risk of investing in this potential PUE activity.

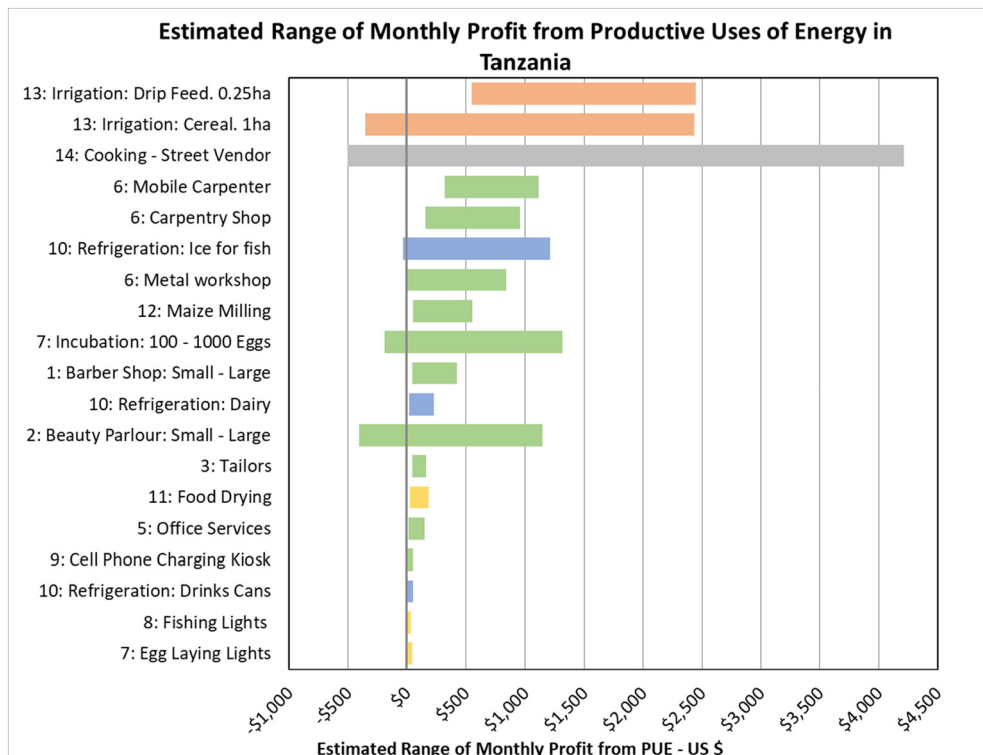


Fig. 6. Estimated range of monthly profit from the PUE applications in Tanzania.

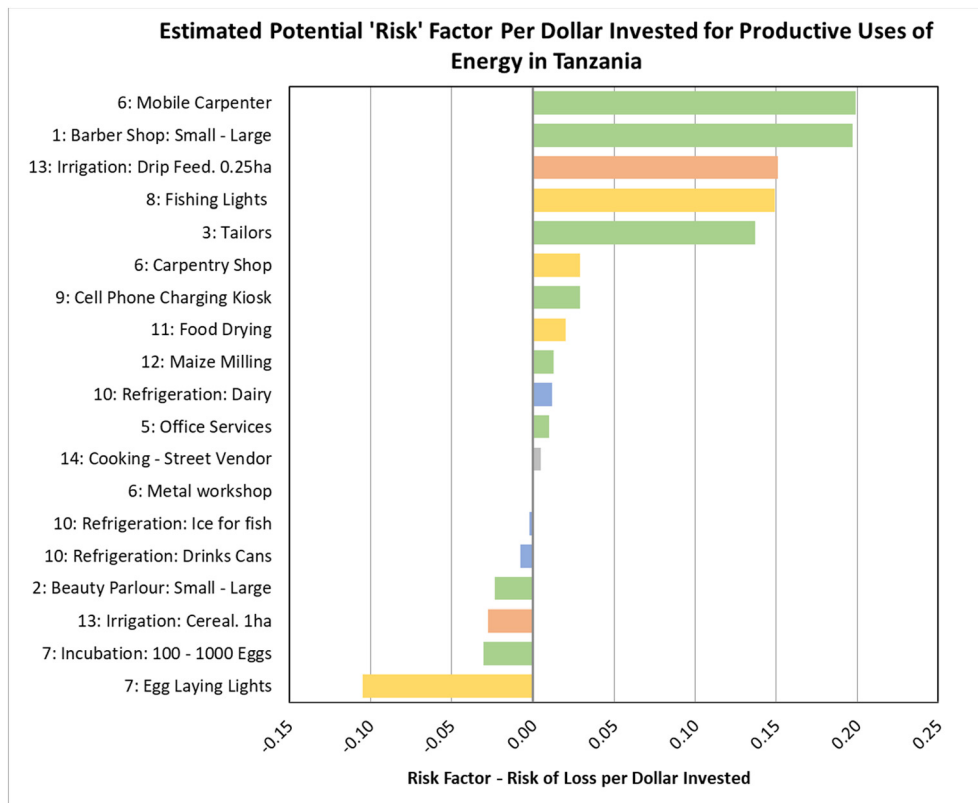


Fig. 7. 'Risk' factor for the PUE applications in Tanzania.

Egg laying lights, for example, are high risk, as this activity is highly dependent upon the sale price of eggs at the market.

Mobile carpenters, barber shops and drip-fed irrigation all are minimal risk for investment and hence development of product and supply chains.

From this analysis in Tanzania, barber shops, mobile carpenters, drip feed irrigation, fishing lights and tailors are highlighted as the PUE systems that show lower investment risk and higher potential monthly profit, with drip feed irrigation having the largest potential market. This compares well with other analysis which shows irrigation systems and barbershops have the highest viability in Malawi (Eales et al., 2017). A Tanzanian barbershop example by GVEP does not show as high potential for income generation, with average profit per dollar invested from \$0.08 - \$0.19 (GVEP, 2015), but this work was from 2015 and PV prices have dropped significantly since then. Access to water has been highlighted as a large barrier to economic productivity in Tanzania (Doumbia et al., 2013), which highlights the potential of drip-feed irrigation systems.

Geographical analysis

Using the data generated by this methodology these PUE systems can be investigated in more geographic detail, with some examples given in Fig. 8, which show the maximum number of PUE systems that could be implemented in each area for some different PUE examples.

These geographical analyses show irrigation systems are widely dispersed, which may make operation and maintenance visits difficult. Barbers are typically identified near to higher population densities, so are potentially have better equipment supply routes. This type of geographical analysis is, potentially, very powerful. System providers and funding organisations can highlight areas for distribution and supply hubs. Areas with a larger potential market for initial trial projects are

quickly and easily highlighted. This paper presents the methodology to produce this type of analysis and can have real value when used for different locations or to compare a new technology or system.

Conclusion

A multi-parameter methodology for evaluating and comparing the technical and economic feasibility of PUE systems has been created and has been successfully used to compare fourteen different PUE applications with 25 different variations. Analysis using this methodology has generated results for the load profiles, the daily energy requirements and, hence, the power system requirements and cost. This has then been used to put values on the potential income generation, variation in potential profit (and hence risk) and the market size for the PUE application. These have been successfully compared, which was the main objective of this work.

The overall recommendations are that barber shops, mobile carpenters, drip feed irrigation, fishing lights and tailors are PUE systems that show the least risk and higher potential monthly profit, with drip feed irrigation having the largest potential market.

This analysis has been an indicative comparison of stand-alone solar PV power systems for different PUE in the specific case of Tanzania. This analysis has been used by the partner organisation to focus on implementing the most viable PUE systems with the best economic rates of return. Threats and opportunities with other PUE systems have also been highlighted. The load profiles created, and the daily energy requirement analysis, may be useful to others performing similar analysis in similar areas.

Further work

For the most viable PUE, with high potential income and lower risk investment further, more detailed, multi-stakeholder market analysis

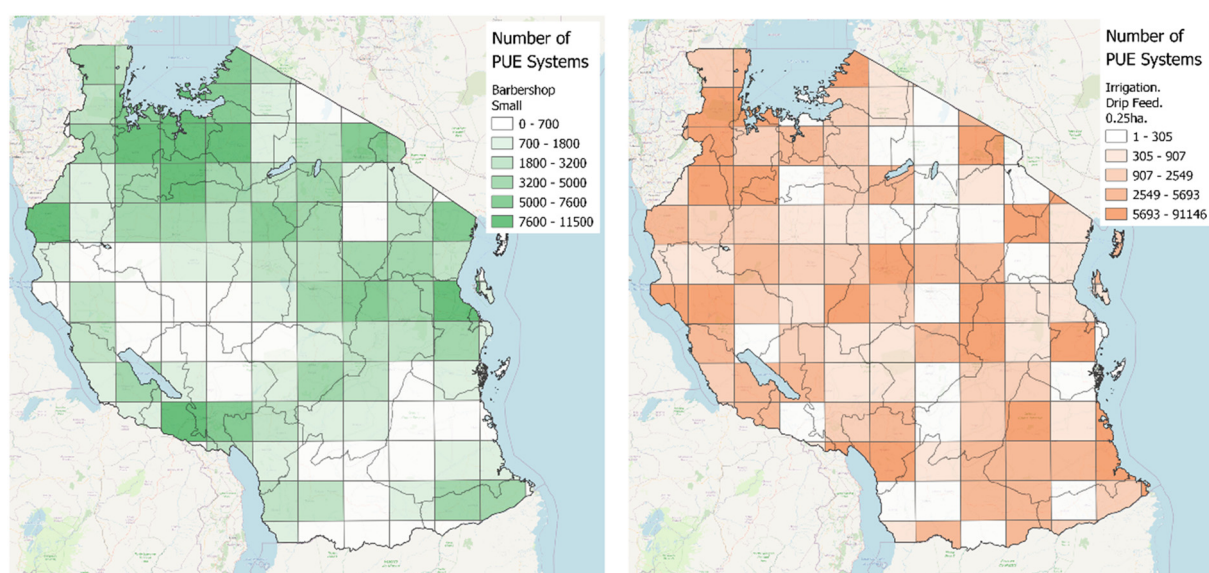


Fig. 8. Total number of small barbershop systems (left) and drip-feed 0.25 ha irrigation systems (right).

is required as the next step. More stakeholders need to be involved in more detailed analysis. This includes suppliers of equipment and finance, government figures, and, most importantly, the potential communities and customers of the PUE and entrepreneurs who will provide the PUE.

The load profiles were generated from estimated use analysis. Real-world measured data for these PUE would be invaluable for confirming the accuracy of the estimated load profiles and hence the power system size requirements.

Further work should include repeating the business surveys to obtain more detailed data from different regions. A business survey will always be required when applying this methodology to other regions.

The values for the bar/café/restaurant systems are not shown on Figs. 6 & 7, as these are add-ons to a PUE activity. The potential income and profit cannot be easily compared with the other activities. Refrigeration for cans of drink is another PUE activity that is difficult to directly compare in this analysis, as the system cost would have to be covered by any premium charged for cold drinks.

Food vendor cooking systems were difficult to analyse with this approach, as the system costs presented are for a solar PV power supply system to provide the full energy requirements throughout the year. With cooking, other sources of fuel can be easily used, so smaller power supply systems could be used. Also, direct solar cooking (with no battery) could be used to further reduce the system cost. Food drying systems were hard to compare to other PUE as they are usually community-led projects with numerous stakeholders. The system cost is primarily hardware cost, with the electrical power supply being a small percentage of the total equipment cost.

More investigation into the ability for people to pay for these services is required. This can vary, including seasonally, and this influences the financing terms and economic models. The reality of 'on the ground' supply and maintenance of equipment requires more investigation and stakeholder mapping, including more detailed costs for transportation and company profit.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.esd.2022.08.018>.

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